

ANSX Vault — Invention Disclosure Summary

For Submission to College Patent / IPR Cell

Title of Invention: Hardware-Instantiated Erasure Coding Apparatus with Split-Authority Codec Instantiation

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Has this invention been publicly disclosed? ☐ Yes ☒ No (Public disclosure includes: published papers, conference presentations, public GitHub repositories, social media posts, product demonstrations to external parties)

1. Problem Statement

Current data security systems protect files using encryption keys — digital passwords that lock and unlock data. The fundamental weakness is that **if the key is stolen, copied, or intercepted, the attacker gains complete access**. Keys can be extracted from memory, intercepted during transmission, or brute-forced.

This invention eliminates the key entirely.

2. What the Invention Does (Non-Technical Summary)

This invention breaks a file into mathematical fragments using Reed-Solomon erasure coding — the same mathematics used in CDs, QR codes, and satellite communications.

Normally, anyone who knows the math rules can reassemble those fragments. This invention **ties the math rules themselves to two physical hardware chips**:

- **Chip 1 (a contactless NFC card):** Determines *which version of the math* is used (the algebraic field structure)
- **Chip 2 (the computer's built-in security chip):** Determines *where in the math* the fragments are placed (the evaluation coordinates)

If either chip is missing or wrong, the math itself produces nonsense. There is no password to steal. The algebra required to rebuild the file does not exist without both physical chips present.

3. What Makes This Novel (Why No One Has Done This Before)

Prior systems use hardware (NFC cards, TPM chips) to **derive a cryptographic key**, then use that key to lock/unlock data processed by a standard, fixed Reed-Solomon codec.

This invention does something fundamentally different: the hardware **defines the algebraic structure of the codec itself**. The Reed-Solomon code literally cannot exist without the hardware. There is no separate key. The hardware IS the math.

Specifically, the novelty is:

What	Prior Art	This Invention
Role of hardware	Generates a key	Defines the algebra

Reed-Solomon codec	Fixed, standard	Changes per hardware pair
What an attacker needs	The key	Both physical chips simultaneously
Key storage	Key exists somewhere	No key exists anywhere

4. How It Works (Technical Summary)

Architecture: Split-Authority Codec Instantiation

The system uses two independent hardware-rooted commitment values:

```
graph TD
    subgraph "First Hardware Root"
        NFC[NFC Integrated Circuit] -->|Immutable EEPROM| C1[Commitment C1]
        C1 -->|HMAC-SHA256 'FIELD'| KF[Field Key KF]
        KF --> Poly[Primitive Polynomial p x]
    end

    subgraph "Second Hardware Root"
        TPM[Host Security Chip / TPM] -->|Platform Identity| C2[Commitment C2]
        C2 -->|HMAC-SHA256 'GEOMETRY'| KG[Geometry Key KG]
        KG --> Eval[N Evaluation Coordinates]
    end

    Poly --> Codec((Reed-Solomon Codec))
    Eval --> Codec

    Codec -->|Reconstruction| Rx[Residual Polynomial R x]
    Rx -->|R x = 0| Success[Hardware Match: Data Reconstructed]
    Rx -->|R x != 0| Fail[Hardware Mismatch: Halt]

    style NFC fill:#d3e4ff,stroke:#0052cc
    style TPM fill:#d3e4ff,stroke:#0052cc
    style Codec fill:#ffe6cc,stroke:#d66a00
```

C₁ (from the NFC card):

- Stored in write-locked, physically immutable EEPROM memory
- Determines the **primitive polynomial** of the Galois Field
- This sets the rules of how all arithmetic (addition, multiplication) works inside the codec
- Uses domain-separated derivation: $K_F = \text{HMAC-SHA256}(C_1, \text{"FIELD"})$

C₂ (from the host computer's security chip):

- Derived from hardware-rooted platform identity
- Determines the **evaluation x-coordinates** of the Reed-Solomon code
- This sets where on the mathematical curve each data fragment sits
- Uses domain-separated derivation: $K_G = \text{HMAC-SHA256}(C_2, \text{"GEOMETRY"})$

Why Both Are Required

Scenario	What Happens
Correct NFC + Correct Computer	Math solves perfectly. File rebuilds.
Wrong NFC card	Wrong field structure. All arithmetic produces wrong symbols.
Wrong computer	Wrong coordinates. Cannot construct the Vandermonde matrix.
No NFC card	Field structure unknown. Codec cannot be instantiated.
No computer chip	Coordinates unknown. Codec cannot be instantiated.

Verification Without Keys

After reconstruction, the system computes a residual polynomial $R(x)$. If $R(x) = 0$, the hardware matched. If $R(x) \neq 0$, the apparatus halts and reports a hardware mismatch. No stored password or key is consulted.

5. Hardware Components Used

Component	Role	Status
NFC integrated circuit with manufacturer originality signature	First hardware root (C_1)	Prototype built
Host platform security chip (TPM or equivalent)	Second hardware root (C_2)	Prototype built
Arduino/PN532 NFC reader	Hardware bridge for NFC communication	Prototype built
Reed-Solomon shatter engine (C++ implementation)	Codec execution	Prototype built
PyQt6 desktop application	User interface	Prototype built

6. Current State of Development

- ☒ Concept / idea stage
- ☒ Working prototype / proof of concept
- ☐ Published paper
- ☐ Product in market

Prototype details: A working desktop application (A.N.Sx Vault) exists with:

- NFC card reading via Arduino hardware bridge
- Reed-Solomon shatter engine (C++ compiled library)
- Cloud shard distribution to GoFile.io
- Blockchain session management via Polygon smart contracts
- Multi-user operator support with hardware lock screen

7. Potential Commercial Applications

Market	Application	Value
Enterprise security	Zero-trust physical file vaults for corporate data	High
Defense / military	Two-person physical data reconstruction without key transmission	Very high
Cryptocurrency	Hardware cold storage without seed phrases	High
Healthcare	Patient data that requires physical device co-presence	Medium
Legal / compliance	Tamper-evident document storage with hardware attestation	Medium

8. Filing Strategy Recommendation

Jurisdiction: India (provisional) → PCT → India + US national phase

Rationale:

- Indian provisional secures global priority date at minimal cost
- PCT provides 30 months before national phase entry
- International Search Report identifies prior art before commitment
- India: apparatus framing defends against Section 3(k)
- US: specific hardware integration defends against Alice/Mayo

Claim structure: 1 independent claim (genus-level) + 7 dependent claims covering specific embodiments

9. Known Prior Art

Reference	What It Covers	Why This Invention Is Different
US6385751B1	Configurable primitive polynomial in RS apparatus	Software-configurable; this invention is hardware-rooted and not software-modifiable
Suh & Devadas 2007	PUF + error correcting codes	Uses hardware for key derivation, not for defining codec algebraic structure
McEliece 1978	Secret code structure as cryptographic primitive	Theoretical; no physical hardware binding or split-authority architecture
NXP NFC authentication patents	NFC originality signatures	Authentication only; not used for codec parameterization

10. Attachments

The following documents accompany this disclosure:

1. **Draft Claims Document** — 1 independent + 7 dependent claims (reviewed by four independent technical reviewers)
2. **Engineering Tasks Document** — Hardware BOM template, co-presence protocol, deterministic mappings

3. **Specification Sections** — Probability analysis, prior art distinction, Section 3(k) defense arguments
4. **Working Prototype** — Available for demonstration